Exercice 3 :

$$y_t = e_t - \theta e_{t-1}$$

The CLSE of θ is :

$S_c(\theta) = \sum_{t=1}^{3} e_t^2$ where conditionally on $e_0 = 0$ we have

$$\begin{cases} e_1 = y_1 = 0 \\ e_2 = y_2 + \theta e_1 = -1 \\ e_3 = y_3 + \theta e_2 = \frac{1}{2} - \theta \end{cases}$$

$$S_c(\theta) = (-1)^2 + \left(\frac{1}{2} - \theta\right)^2$$

$$S_c(\theta) = 1 + \left(\frac{1}{2} - \theta\right)^2$$

$\hat{\theta}$ verify $\frac{\partial S_c(\hat{\theta})}{\partial \theta} = 0$

$$\iff \boxed{\hat{\theta} = \frac{1}{2}}$$

for an MA(1), $\hat{\gamma}_0 = \hat{\sigma}_e^2(1 + \hat{\theta}^2) = \frac{1}{3-1}\left[0^2 + (-1)^2 + \left(\frac{1}{2}\right)^2\right]$

$$= \frac{5}{12}$$

so the MME estimator of σ_e^2 is : $\hat{\sigma}_e^2 = \frac{5}{12\left(1 + \left(\frac{1}{2}\right)^2\right)}$ $\boxed{\hat{\sigma}_e^2 = \frac{1}{3}}$